ABUBAKR M. ABDELHALIM

Electrical and Electronic Engineering Student | **Universiti Teknologi Malaysia**
Skudai, Johor, Malaysia | +60-123526371
abdelhalimabubakr@gmail.com | **LinkedIn Profile**



Summary

Third-year **Electrical Engineering** student at **UTM** specializing in **robotics and control systems**. Experienced **Electronics Head of Department** for the national team representing Malaysia in international competitions. Proven expertise in **4-layer PCB design**, **industrial automation**, and **embedded systems**.

Education

Universiti Teknologi Malaysia (UTM) Johor, Malaysia
Bachelor of Engineering: Electrical and Electronic Engineering Mar 2023 – Current

- **Leadership: Electronics Head of Department** at **UTM Robocon club**.
- **Professional Membership:** Student Member – **Saudi Council of Engineers**.

Experience

Electronics Head of Department (UTM Robocon Team) Jan 2024 – Apr 2025
Universiti Teknologi Malaysia (Representing Malaysia) Johor, Malaysia

- **National Representation:** Led the electronics division for the official UTM team representing Malaysia in international robotics competitions (**ABU ROBOCON** 2024 & 2025).
- **Technical Project:** Spearheaded electrical design for high-speed shooters using **STM32** and **PID Control**; fabricated internal **PCBs** for belt synchronization and energy release.
- **Systems Engineering:** Designed a **4-layer main control boards** (STM32F4) and precision motor interface boards (STM32G4) with **CAN communication**.
- **Leadership:** Managed cross-functional integration between mechanical and software teams; mentored 20+ members in **Altium Designer** and **SolidWorks**.
- **Foundations:** Developed autonomous line-follower robots and mastered **PCB prototyping** under tight deadlines as a Junior Member.

Part-Time Engineer Apr 2025 – Jul 2025
TSH Group Johor, Malaysia

- **Industrial Automation:** Mastered **PLC testing** and industrial **wiring diagrams** for factory automation systems.
- **Quality Control:** Authored **Work Instructions (WI)** for MTC machine testing, reducing industrial hardware defects.
- **Prototyping:** Designed and built a custom **DI/DO testing kit** for enhanced signal verification in the field.

Engineering Projects Portfolio

- **Self-Balancing Robot:** Developed a two-wheel system using **ESP32** and **MPU6050**; tuned real-time **PID loops** for stability and disturbance recovery.
- **Autonomous Navigation:** Built a line-follower car with **Bluetooth control** using **STM32**, focusing on **sensor fusion** and efficient motor routines.
- **Hardware Solutions:** Designed and fabricated custom **PCBs**, including **Isolated Valve Drivers**, **DC-DC Buck Converters**, **ESP32 breakout boards**, and **STM32 development boards**.

Leadership & Involvement

- **Organizing Committee Member: IYRC 2025** (International Youth Robot Competition). Coordinated technical logistics and promoted international engineering engagement.
- **Technical Workshop Lead:** Conducted hands-on workshops on **embedded systems**, **PCB design**, and **microcontroller programming** for undergraduate students.

Technical Skills

- **Programming: C/C++**, **Python**, **MATLAB**, **Arduino**, **Embedded C**
- **Embedded & Hardware: STM32**, **ESP32**, **PLC Testing**, **Industrial Wiring**, **4-Layer PCB Design**
- **Control & Tools: Altium Designer**, **KiCAD**, **SolidWorks**, **Proteus**, **VS Code**, **STM32CubeIDE**, **Simulink**
- **Professional: Project Management**, **Technical Mentorship**, **Cross-Functional Leadership**